Phase I Swindon EV

Geo-Environmental
Phase 1 Desk Study

Swindon EDA Consortium

Brookbanks

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Historical site mapping – Ordnance Survey

Site Investigation Report

Site Investigation Technical Note

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1 Introduction

1.1 Brookbanks Consulting Ltd is appointed by the Swindon EDA Consortium to complete a Phase 1 Geo-Environmental Desk Study for the proposed residential development consisting Phase I of the Swindon EV element of the wider EDA.

1.2 The objective of the study is to demonstrate that the development proposals are acceptable from a geotechnical and chemical viewpoint.

2 Background Information

Location & Details

2.1 The proposed development lies to the east of Swindon. The A420 and the Great Western Rail Line bound the site to the south. To the west of the site, Thornhill Lane forms the site boundary, with the extents of South Marston, Nightingale Lane and the accompanying farmland bounding the site to the north. Agricultural land forms the eastern boundary of the site, beyond which is the River Cole.

2.2 The majority of the land is thought to be historically undeveloped, the site location and boundary are shown indicatively on Figure 2a, below:

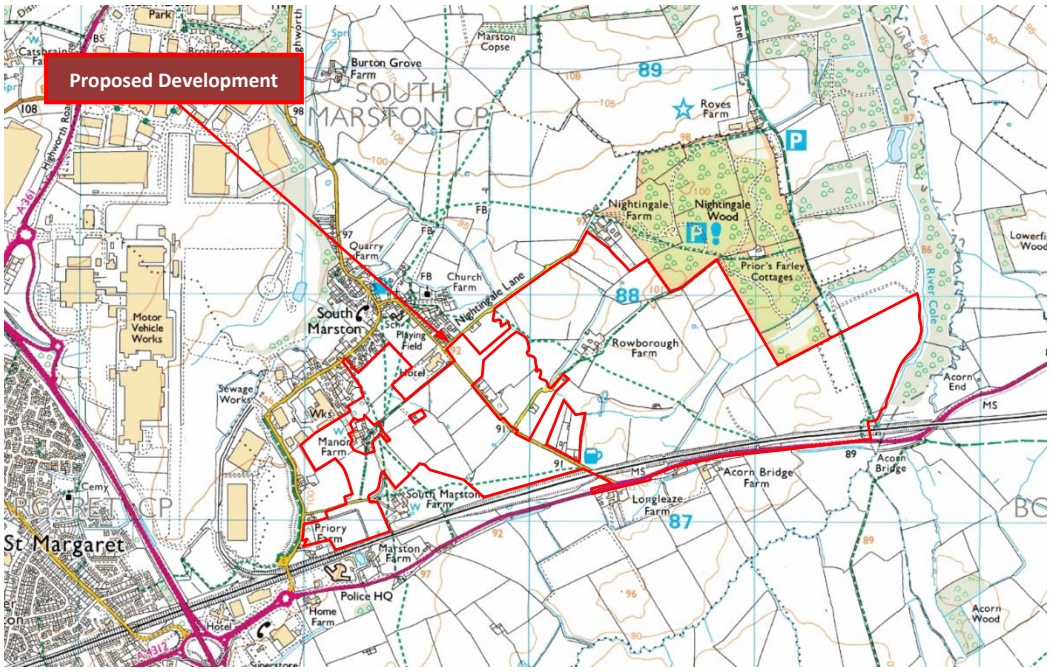


Figure 2a: Site location

Development Criteria

2.3 The following development profile is proposed within the 165 ha site:

- Up to 2,380 residential units
- A Neighbourhood Mixed Use Centre comprising up to 1,500m^2 of A1 Retail and up to 1,500m^2 of B1 employment, A2, A3, A4, A5, C3, D1 and D2 Uses
- Primary School

Sources of Information

2.4 The following bodies were consulted during the study:

- Environmental Matters - Environment Agency

2.5 The following information has been gathered during the study:

- Environmental Search - Envirocheck Report, July 2008
- Geo-Environmental Investigation Report - Geotechnical Developments (UK) Ltd, January 2009
- Published Geology - British Geological Survey

3 Historical Site Uses

3.1 In appraising the site history, published Ordnance Survey maps have been reviewed dating from the late 19th century up to the present day. A list of historical maps used in this report is contained within the Appendix along with a selection of large scale maps for reference.

3.2 Inspection of the Ordnance Survey maps has revealed the following information relating to the former uses of the site and immediate environs.

Pre 1900 The site comprises of numerous large agricultural fields with the northern and eastern boundaries confined by further fields. The south of the site is bound by the Great Western **Railway line** that passes from east to west towards the Swindon Centre. The western boundary is bound by further fields. There are numerous farm buildings and cottages identified within the site boundary, including a vicarage on the north western boundary. A road runs from the South Marston settlement circa 200m north west of the boundary down to the main road running parallel to the south of the Railway line. The **South Marston Brook** is seen to run through the centre, entering the site from the north west and leaving to the east forming a confluence with the **River Cole**. The Wiltshire and Berkshire **Canal** is also located to the south of the railway line.

1925 There are no significant changes reported on site. However, the canal to the south of the site has now been listed as disused. An **old quarry** has identified circa 650m north west of the site.

1960 No changes of significance to report on site. The road running parallel to the railway line and north east of the site has now been named the **A420**, the disused canal has been renamed the old canal and the old quarry previously noted is no longer present.

1978 – 81 No significant changes have been identified on the site. The South Marston settlement has undergone minor expansion with new **allotment gardens**, a **quarry farm** and further dwellings reported. New **engineering works** and **sewage works** have also been located circa 100m north west of the site. To the south west of the site a new major road has been constructed, with which two large housing developments have been developed. A section of the old canal can longer be observed, which suggests that this may have been **infilled**.

1999 There are no significant changes on site. The South Marston area has continued to develop with further residential dwellings and the expansion of school grounds, the vicarage has been converted into a large hotel. The location of the engineering works has expanded into an **industrial estate** with further **works** identified adjacent to the sewage works. The south west of the site has continued to develop with the

addition of a commercial area circa 100m from the boundary, this includes two new **superstores**. Parcels of new woodland have been identified to the north, close to the boundary.

2008 – 2012 Some minor growth of developments within the site boundary have been identified including along the road now named as Old Vicarage Lane. The South Marston area has expanded further, along with the woodland previously reported. The sewage works to the north west are now disused, and further west circa 250m from the boundary, two **large warehouses** have been erected with the addition of an **electricity substation**. South of the site, between the railway line and A420, new police headquarters have been developed including a large drainage basin adjacent.

4 Recent & Current Site Usage

4.1 The majority of the site is currently undeveloped. The historical map search suggests that the land has previously been set as agricultural land.

5 Ground Conditions

Geology

5.1 With reference to the British Geological Survey maps, a number of different geological strata are shown across the site. To the south and west the site is reported to be predominantly underlain by mudstone belonging to the Ampthill Clay Formation and Kimmeridge Clay Formation with an outcrop of Ferruginous sandstone in the western corner and south belonging to the Down Sand Member.

5.2 To the north and east, the BGS mapping shows the general ground conditions to consist a mix of sandstone to the north and north-east (Hazelbury Bryan Formation and Kingston Formation) and limestone (Stanford Formation) to the east. The south-eastern corner is underlain by the Ampthill Clay Formation and Kimmeridge Clay Formation.

5.3 With regards to superficial deposits, these are shown to be limited to a thin band of alluvium (clay, silt, sand and gravel) passing through the site, generally along the line of the South Marston Brook. The published geology is shown in Figure 2b, below:

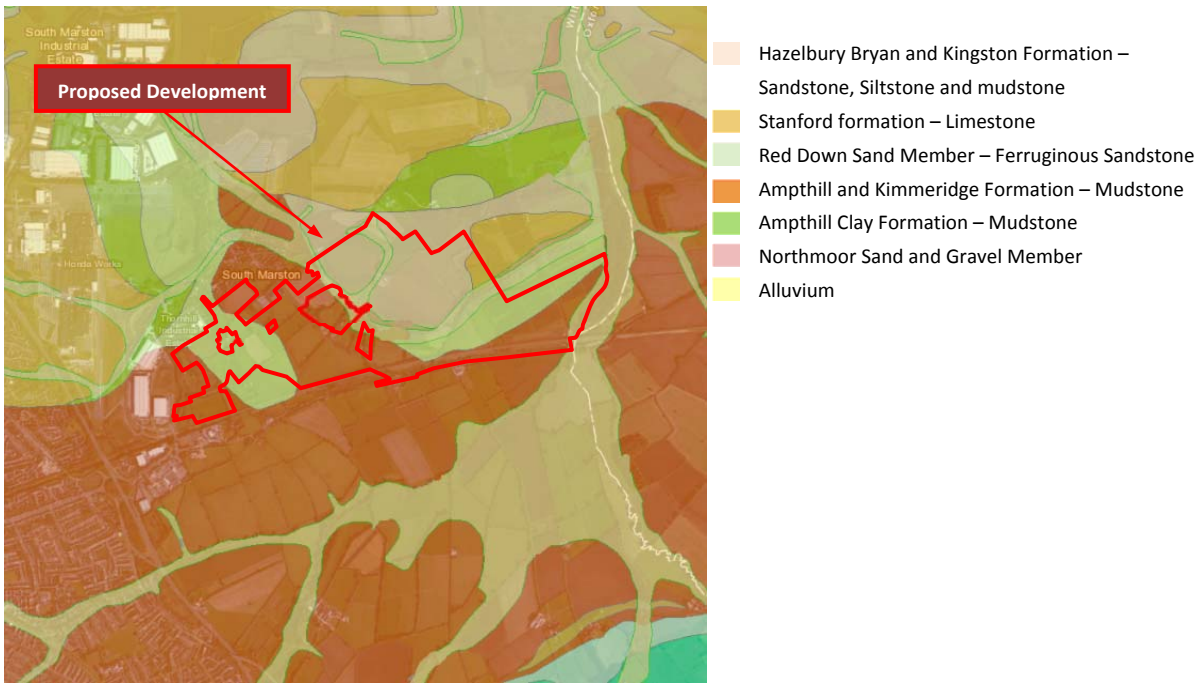


Figure 5a: BGS Published Geology

5.4 Site specific investigations across the wider site have yet to be completed however some localised borehole records have been obtained which generally confirm the ground conditions to be as per the BGS records.

5.5 BGS records within the Envirocheck Report indicate the following ground stability hazards:

- Collapsible ground stability No Hazard*
- Compressible ground: No Hazard*/ Moderate*
- Ground dissolution: Very Low*/ Low*
- Landslide: Very Low*/ Low*
- Running sand: No Hazard* / Very Low*/ Low*
- Shrinking & Swelling: No Hazard*/ Very Low*/ Moderate*
- Shallow mining No Hazard*

(* Indicates hazard on site or within 250m of the boundary)

Radon

5.6 Parts of the site are identified to lie within a radon affected area with between 1% and 3% of homes lying above the action level.

5.7 However, it has been reported that no radon protection measures are necessary in the construction of new dwellings or extensions.

Mining

5.8 The site lies on an area that is not reported to be affected by coal mining.

6 Hydrology

Flooding

6.1 The Environment Agency's (EA) National Generalised Modelling (NGM) Flood Zones Plan indicates predicted flood envelopes of Main Rivers across the UK. In many circumstances, the NGM is based on basic catchment characteristic data and broad scale modelling techniques. While this information may be appropriate for general land use planning, it is often necessary to prepare more detailed hydrology, survey and computational models where proposed developments lie adjacent to a watercourse. Figure 6a, below shows the EA's Flood Zone mapping of the area.

6.2 The Flood Zone plan for the area shows significant out of bank flows along the South Marston Brook which runs through the centre of the site and along the River Cole to the east. Flows on these watercourses are seen to come out of bank during the 1 in 100 (1% AEP) and 1 in 1,000 year (0.1% AEP) events, with the corresponding floodplains covering some residential dwellings within the site.

6.3 The mapping shows that the site lies within Flood Zones 1, 2 and 3; being an area of High Probability of flooding, lying within the catchments of both the 1 in 100 (1% AEP) and 1 in 1,000 (0.1% AEP) year flood events.

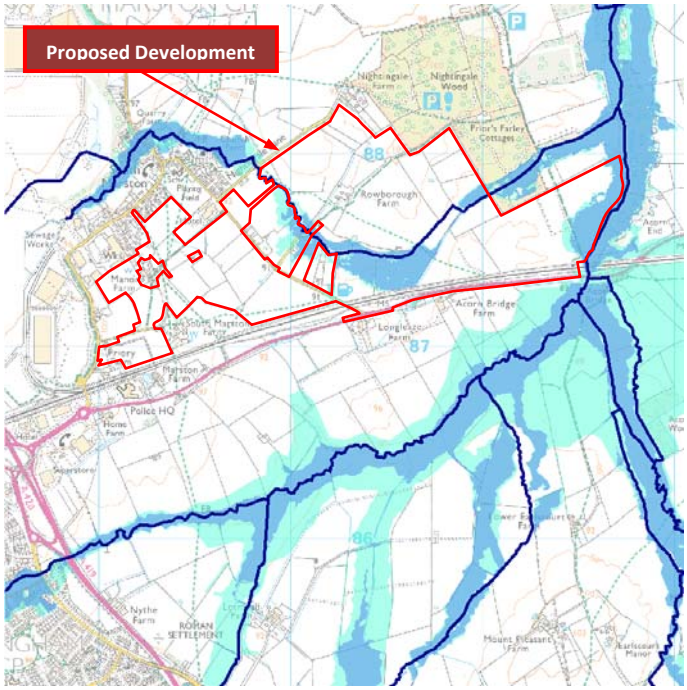


Figure 6a: EA Flood Zone Plan showing 1 in 100 & 1 in 1,000 year floodplains

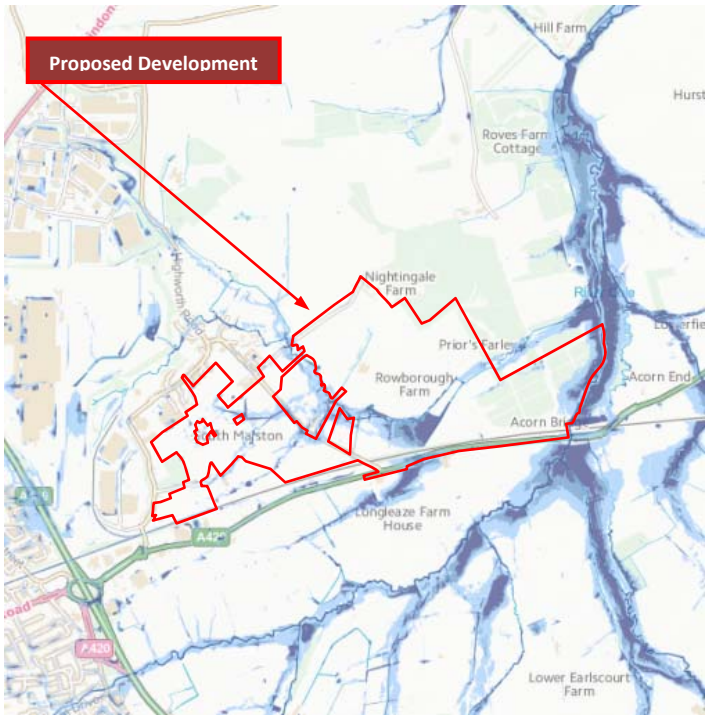
Flooding from rivers without defences – 1 in 100 year (1%) event (Zone 3)

Extent of extreme flood – 1 in 1,000 year (0.1%) event (Zone 2)

Flood defences

Areas benefiting from flood defences

6.4 Advice provided by the Government's online Flood Warning Information Service, indicates that the overall risk of surface water flooding on site is outlined below in Figure



High Risk

Medium Risk

Low Risk

Very Low Risk

Figure 6b: EA Risk of Flooding from Surface water

Discharge Consents

6.5 Three discharge consents have been granted within the proposed site, these are indicated below in Table 6c:

Operator	Date	Discharge Type	Receiving Water	Distance (m)	Direction
Thames Water Utilities	Nov 1989	Sewage Discharge/ Pumping Station	South Marston Brook	0	NW
Thames Water Utilities	Nov 1989	Sewage Discharges/ Pumping Station	South Marston Brook	0	W
E H Wildern	Mar 1971	Other Matter/ Fish Farm	Tributary of South Marston Brook	0	N

Table 6c: Discharge Consent

6.6 No further discharge consents have been granted within 250m of the site boundary.

Water Quality

6.7 The Environment Agency currently monitor 40,000km of rivers across England. To help protect these areas each stretch of river is monitored and given a river quality grade. This is based upon the chemical quality of the water. The rivers are then graded from A to E with A representing a river with very good water quality and E, a river with very poor water quality.

6.8 Two river quality sampling points have been recorded within the site; one in the centre along South Marston Brook, and another to the south east along the River Cole, these were last sampled in 2000 and indicated a Grade B 'Good' for water quality.

6.9 To improve the quality of water bodies, European legislation known as the Water Framework Directive (WFD) was introduced in 2009 to promote a new approach to water management through river basin planning. One aim of the Water Framework Directive was to improve the ecological health of inland and coastal waters and to prevent further deterioration. A requirement was placed on nearly all inland and coastal waters to achieve 'Good' status by 2015. However, for many local authorities this has been extended until 2027.

7 Hydrogeology

Aquifer Designation

7.1 According to the Environment Agency the underlying geology on site is shown to comprise Secondary A Aquifers and Unproductive Strata.

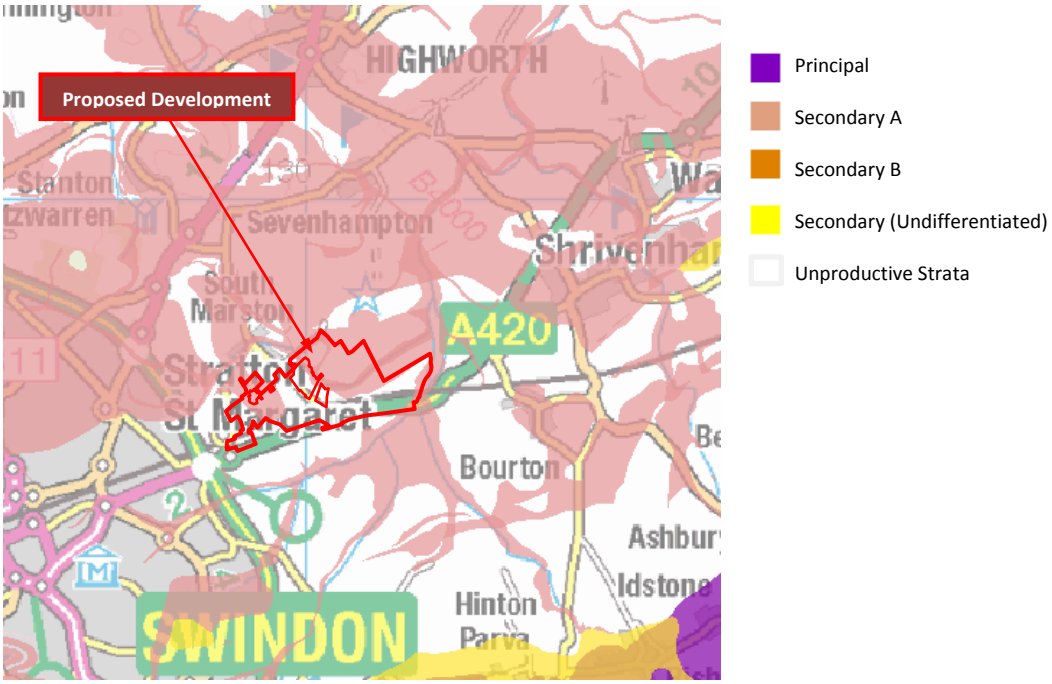


Figure 7a: BGS Designated Aquifers

7.2 According to the Environment Agency the Superficial deposits on site is shown to comprise Secondary A Aquifers and Unproductive Strata.

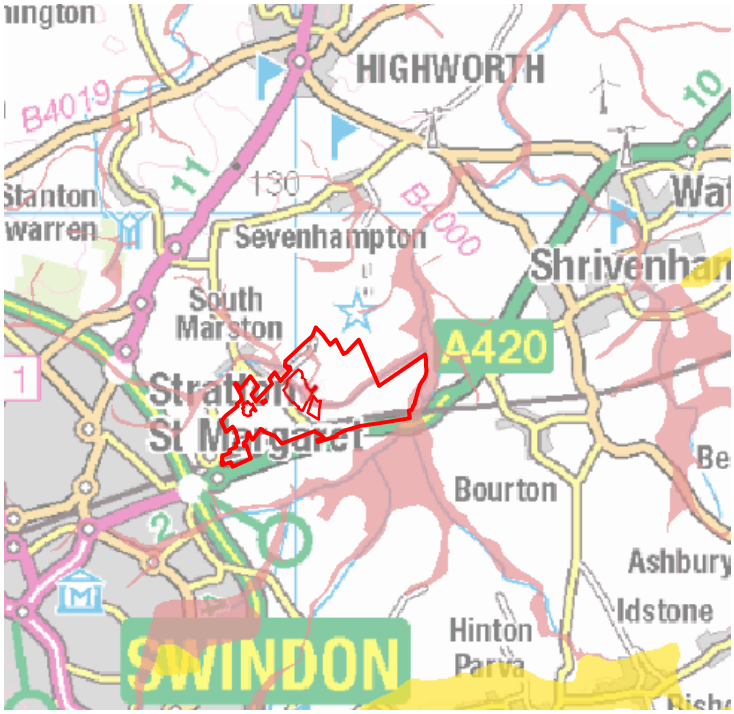


Figure 7b: EA Superficial Deposits Designation

7.3 According to the Environment Agency the Superficial deposits on site is shown to comprise Secondary A Aquifers and Unproductive Strata.

Groundwater Vulnerability Zones

7.4 In terms of groundwater vulnerability zones the majority of the site is reported to be underlain by a minor aquifer with soils of intermediate leaching potential. Areas in the south-western corner, south-eastern corner and in the centre of the site are shown to form non aquifers and the furthest south-east shown to form Minor Aquifer with soils of high leaching potential. This is illustrated below on Figure 7c.

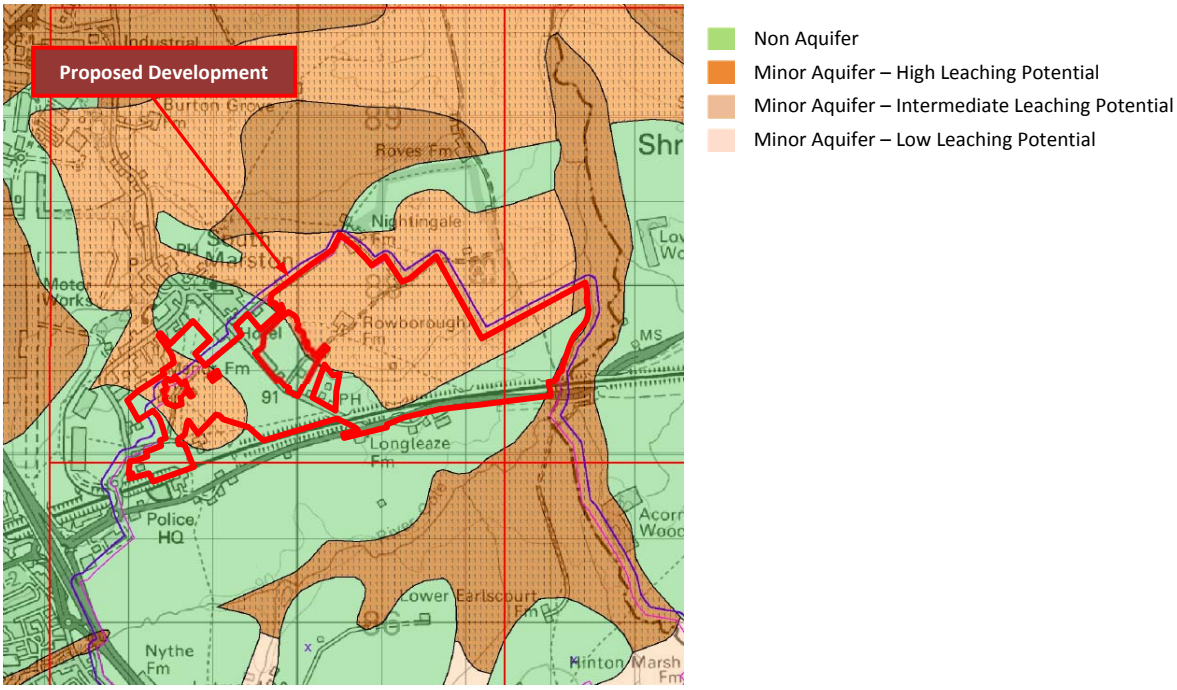


Figure 7c: Groundwater Vulnerability

7.5 Non Aquifers are formations of negligible permeability, usually regarded to contain insufficient quantities of groundwater. However, groundwater flow through non aquifers is a possibility even if a low one.

7.6 Minor Aquifers are fractured or potentially fractured rocks, which do not have a high primary permeability, or other formations of variable permeability including unconsolidated deposits. Although not producing large quantities of water for abstraction, they are important for local supplies and in supplying base flow to rivers.

Groundwater Abstractions

7.7 One ground water abstraction license has been granted within the site boundary, this has been detailed below in Table 7d:

Abstractor/Company	Abstraction	Distance From Site (m)	Direction	Date
Mr R G Wildern	Farming and Domestic	On site	N	1991

Figure 7d: Ground water extraction licences

7.8 No other ground water abstraction licences have been granted within 250m of the site boundary.

7.9 There are no source protection zones within 1,000m of the site boundary.

8 Potential Contaminative Uses & Statutory Registers

8.1 Two **Pollution Incidents to Controlled Waters** have been recorded within the site with another identified within 50m of the site boundary, these have been indicated below in Table 8a:

Pollutant	Receiving Water Body	Category	Distance (m)	Direction
Unknown Sewage	Not given	3 – Minor Incident	On site	SW
Unknown Sewage	Not given	3 – Minor Incident	On site	W
Unknown	Not given	3 – Minor Incident	42	SE

Table 8a: Pollution Incidents to Controlled Waters

8.2 One **Substantiated Pollution Incident Register** has been identified in the south west of the site, this is in relation to a Category 2 (significant) land impact caused by a general biodegradable pollutant.

8.3 One **Historic Landfill Site** has been recorded on site, the details of which are as follows in Table 8b:

Name	Last Input	Waste Type	Distance (m)	Direction
Roger Stuart Chrisp, Acorn Bridge	1987	Deposited/Inert waste	On site	E

Table 8b: Historic Landfill Sites

8.4 One **Local Authority Recorded Landfill** has also been reported on site. The following information has been provided in Table 8c:

Name	Activity	Waste Type	Distance (m)	Direction
Crappers and Sons, Acorn Bridge	**Closed**	Soil, Subsoil	On site	SE

Table 8c: Local Authority Recorded Landfill

8.5 None of the following have been recorded within 1,000m of the site boundary:

- Contaminated land Register Entries and Notices
- Enforcement and Prohibition Notices

- Integrated Pollution Controls
- Integrated Pollution Prevention and Control
- Local Authority Integrated Pollution Prevention And Control
- Local Authority Pollution Prevention and Control Enforcements
- Prosecutions Relating to Authorised Processes
- Prosecutions Relating to Controlled Waters
- Registered Radioactive Substances

Hazardous Substances

8.6 There are no records of the following on or within a 1,000m radius of the site boundary;

- Control of Major Accident Hazards Sites (COMAH)
- Explosive Sites
- Notification of Installations Handling Hazardous Substances (NIHHS)
- Planning Hazardous Substance Consents
- Planning Hazardous Substance Enforcements

8.7 There are two recorded **Contemporary Trade Directory Entries** within 250m of the site boundary which commonly relate to:

- Road Haulage Services
- Garage Services

8.8 Further to this, numerous more are identified within 2000m of the site boundary generally relating to:

- Car Dealers and Manufacturers
- Printers
- Builders Merchants
- Paint Manufacturers
- Distribution Services
- Furniture repairs
- Car dealers
- Domestic appliances

8.9 No **Fuel Station Entries** have been recorded within 250m of the site boundary.

Waste

8.10 Acorn Bridge, a **Historical Landfill Site**, is reported in the east of the site (as illustrated on Figure 11a) and was licensed to Roger Stuart Chrisp of Chrisp Transport. Deposited waste included inert waste and the site is reported to have been active between June 1977 and December 1987.

8.11 One **Local Authority Recorded Landfill Site** is reported within 1,000m of the site boundary by Wiltshire County Council. The site, now closed, is reported to have been situated in the south east of the site and was operated by Crappers and Sons Limited at Acorn Bridge. Waste at the site included soil and subsoil deposits.

8.12 Swindon Unitary Council is reported to have had no landfill data available to supply.

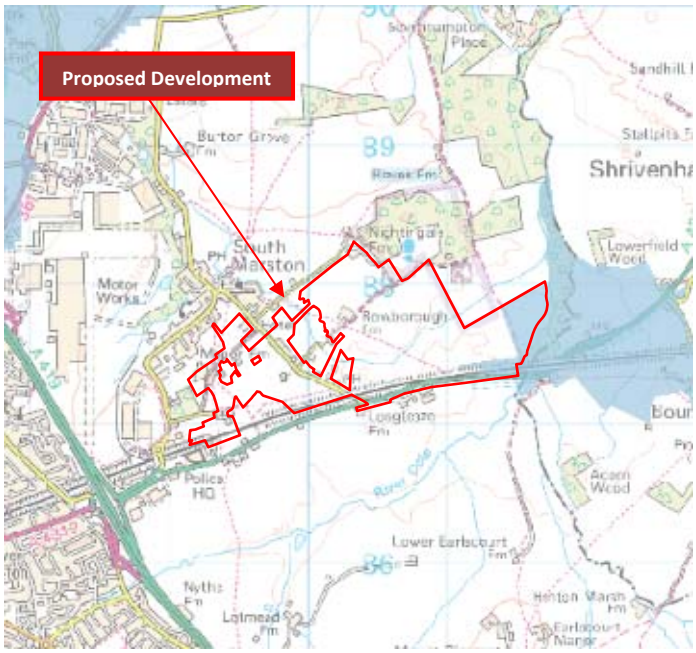
8.13 There are no provided reports of the following within 1,000m of the site boundary:

- BGS Recorded Landfill Sites
- Integrated Pollution Control Registered Waste Sites
- Licenced Waste Management Facilities (Landfill Boundaries)

- Licenced Waste Management Facilities (Locations)
- Registered Landfill Sites
- Registered Waste Transfer Sites
- Registered Waste Treatment or Disposal Sites

9 Environmental Setting

9.1 A small area in the south east of the site is shown to lie within a surface water **Nitrate Vulnerable Zone** as illustrated below in Figure 9a:



Surface water Nitrate Vulnerable Zone

Figure 9a: Nitrate Vulnerable Zones

9.2 None of the following are reported within 1000m of the site boundary:

- Ramsar Sites
- Area of Outstanding Natural Beauty
- Sites of Specific Scientific Interest
- Special Areas of Conservation
- Special Protection Areas
- Areas of Adopted Greenbelt
- Areas of Unadopted Green Belt
- Environmentally Sensitive Areas
- Forest Parks
- Marine Nature Reserves
- National Nature Reserves
- National Parks
- Nitrate Sensitive Areas

10 Site Conceptual Model

10.1 Guidance has been published by the Department of the Environment, Transport and the Regions (DETR Circular 02/2000) 'Environmental Protection Act 1990: Part 11A – Contaminated Land (20th March 2000) which promotes the 'suitable for use approach'. This has since been replaced by the DEFRA: Contaminated Land Statutory Guidance (April 2012). The DEFRA guidance note focuses on the risks caused by land contamination. The approach recognises that the risks presented by any given level of contamination will vary greatly according to the use of the land and a wide range of other factors, such as the underlying geology of the site. Risks therefore need to be assessed on a site-by-site basis.

10.2 The "suitable for use" approach consists of three elements:

- Ensuring that land is suitable for its current use - in other words, identifying land where contamination is causing unacceptable risks to human health and the environment, assessed on the basis of the current use and circumstances of the land, and returning such land to a condition where such risks no longer arise ("remediating" the land): the new contaminated land regime provides general machinery to achieve same.

- Ensuring that land is made suitable for any new use, before planning permission is given for that new use - in other words, assessing the potential risks from contamination, on the basis of the proposed future use and circumstances, prior to planning permission being granted for the development and, where necessary to avoid unacceptable risk to human health and the environment, remediating the land before the new use commences; Ensuring land is appropriately remediated prior to any new use is commenced is dealt with under the planning process, and planning conditions such as pre-construction remediation can be applied to this effect.

- Limiting requirements for remediation to the work necessary to prevent unacceptable risks to human health or the environment in relation to the current use or future use of the land for which planning permission is being sought - in other words, recognising that the risks from contaminated land can be satisfactorily assessed only in the context of specific uses of the land (whether current or proposed), and that any attempt to guess what might be needed at some time in the future for other uses is likely to result either in premature work (thereby risking distorting social, economic and environmental priorities) or in unnecessary work (thereby wasting resources).

10.3 Also addressed within the DEFRA guidance is the issue of 'contaminated land'. The DEFRA guidance explains that before the **Local Authority** can make the judgement that any land appears to be **contaminated land** on the basis that s**ignificant harm** is being caused, or that there is a **significant possibility** of such harm being caused, the authority must therefore identify a **significant pollutant Linkage**.

10.4 This means that each of the following has been identified:

- A Contaminant Source
- A Pathway
- A Receptor

(This is known as the source-pathway-receptor (SPR)

and that:

- The **contaminant** is causing **significant harm** to **that receptor.**

Or

- There is a **Significant Possibility** of such harm being caused by the **Contaminant to the Receptor**.

10.5 Where any of the three elements of the source-pathway-receptor (SPR) are not present, there is no risk and therefore land cannot be classified as statutory 'contaminated land'.

10.6 In terms of controlled waters, DEFRA: Contaminated Land Statutory Guidance (April 2012) notes the following:

"A.35 Section 78A (9) defines the pollution of controlled waters as: 'The entry into controlled waters of any poisonous, noxious or polluting matter or any solid waste matter'.

A.36 Before determining that pollution of controlled waters is being, or is likely to be, caused, the local authority should be satisfied that a substance is continuing to enter controlled waters or is likely to enter controlled waters. For this purpose, the local authority should regard something as being "likely" when they judge it more likely than not to occur.

A.37 Land should not be designated as contaminated land where:

(a) A substance is already present in controlled waters;

(b) Entry into controlled waters of that substance from land has ceased; and

(c) It is not likely that further entry will take place.

A.38 Substances should be regarded as having entered controlled waters where:

(a) They are dissolved or suspended in those waters; or

(b) If they are immiscible with water they have direct contact with those waters on or beneath the surface of the water.

A.39 The term "continuing to enter" should be taken to mean any entry additional to any which has already occurred."

10.7 In 2004 the EA published the 'Model Procedures for the Management of Land Contamination', CLR11, which provides the technical framework for applying a risk management process, based on the 'suitable for use' approach, when dealing with land affected by contamination.

10.8 In 2008, to enable the practical application of good practice of the EA's Model Procedures CLR11, R&D Publication 66 'Guidance for the Safe Development of Housing on Land Affected by Contamination' was published by the National House Builders Council (NHBC), the EA and the Chartered Institute of Environmental Health. Whilst written to be relevant to housing development it is also applicable to other forms of development where sites are land affected by contamination. The guidance describes in detail the process and activities involved for the identification and assessment of hazards for a Phase 1 assessment.

10.9 At Phase 1 stage, it is necessary to develop an initial conceptual site model to understand the possible relationships between contaminants, pathways and receptors (SPR linkages). If a hazardous source, via an exposure pathway to a potential receptor can be established then there is a 'pollutant linkage', which is preliminarily risk assessed using parameters summarised in Table 10a, below. At this stage, the conceptual model is prepared without site specific soils, groundwater or gas testing and as such, the findings should be treated only as first and general indications of possible SPR linkages.

10.10 The primary potential sources of contamination at the site are indicated below, and shown on Figure 10a:

On-site:	**Agricultural Use**	-	Soil and Water Contamination (Source 1)
	Historic landfill Site	-	Soil and Water Contamination (Source 2)
Off-site:	**Agricultural Use**	-	Soil and Water Contamination (Source 3)
	Railway Lines	-	Soil and Water Contamination (Source 4)
	Ind. Estates	-	Soil and Water Contamination (Source 5)

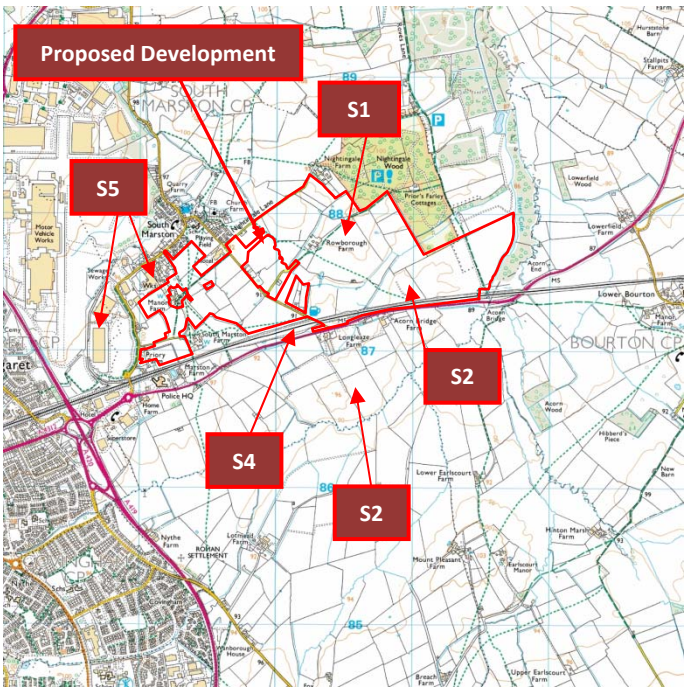


Figure 10a: Potential sources of Contamination

10.11 The potential receptors at the site are:

- End users / site occupiers
- Adjacent users / occupiers
- Controlled waters
- Flora and fauna
- Buildings & construction materials

10.12 The potential pathways at the site are primarily:

- Direct ingestion of soil / water / fruit or vegetable
- Inhalation of dust / vapours
- Direct skin contact with the ground / water
- Regression of plant growth due to phytotoxic contamination
- Vertical and lateral migration of contamination

10.13 An initial methodology has been developed to help identify the potential contamination risk and linkages. The severity of damaging effects and the likelihood of any linkage have been considered.

10.14 Given the potential consequence and likelihood, a risk rating is given, based on the following matrix.

		Consequence			
		Severe	Moderate	Mild	Minor
Probability (Likelihood)	Highly Likely	Very High	High	Medium	Low
	Likely	High	Medium	Medium/Low	Low
	Possible	Medium	Medium/Low	Low	Very Low
	Unlikely	Medium/Low	Low	Very Low	Very Low

Table 10b: Risk ratings

10.15 The risk ratings are described as follows:

Very High: There is a high probability that severe harm could arise to a designated receptor from an identified hazard at the site without appropriate remediation action.

High: Harm is likely to arise to a designated receptor from an identified hazard at the site without appropriate remediation action.

Medium: It is possible that without appropriate remediation action harm could arise to a designated receptor. It is relatively unlikely that any such harm would be severe, and if any harm were to occur it is more likely that such harm would be relatively mild.

Low: It is possible that harm could arise to a designated receptor from an identified hazard. It is likely that, at worst if any harm was realised any effects would be mild.

Very Low: The presence of an identified hazard does not give rise to the potential to cause harm to a designated receptor.

Source	Pathway	Receptor	Comment	Risk Rating	Potential Mitigation
Contaminated soils **On-site:** • Agricultural • Landfill **Off-site:** • Agricultural • Railway line • Ind. Estates	Direct Ingestion & contact	Site workers & occupiers	The majority of the site is historically undeveloped. Agricultural use may have included the use of pesticides and fertilizers which pose a minor potential contamination threat.	Very Low	-
	Inhalation of dust		The majority of the site is historically undeveloped. On-site land uses do not suggest a significant potential for contamination.	Very Low	-
	Direct skin contact		The majority of the site is historically undeveloped. Agricultural use may have included the use of pesticides and fertilizers which pose a minor potential contamination threat.	Very Low	-
	Vertical & lateral migration	Controlled waters	Site is situated on a both Non Aquifer and Minor Aquifer. Site not within ground water protection zone. Historically undeveloped site. Off site sources have the potential to impact upon the proposed site however the site is situated on a high point for the area and as such, migration onto the site is not considered a high risk.	Low	-
	Direct uptake	Flora	The majority of the site is historically undeveloped. On-site use encourages plant growth.	Very Low	-
	Direct contact	Building materials	The majority of the site is historically undeveloped. Agricultural land uses are not considered to have a detrimental impact on building materials.	Very Low	-
Contaminated Groundwater **On-site:** • Agricultural • Landfill **Off-site:** • Agricultural • Railway Lines • Ind. Estates	Direct Ingestion & contact	Site workers & occupiers	Site is situated on both a Non Aquifer and a Minor Aquifer. Groundwater flow into site is a very low possibility however some of the surrounding off-site sources are hydraulically upstream of the site. The majority of the site is historically undeveloped.	Low	-
	Direct skin contact		Site is situated on both a Non Aquifer and a Minor Aquifer. Groundwater flow into site is a very low possibility however some of the surrounding off-site sources are hydraulically upstream of the site. The majority of the site is historically undeveloped.	Low	-
	Vertical & lateral migration	Controlled waters	Site is situated on both a Non Aquifer and a Minor Aquifer. Groundwater flow into site is a very low possibility however some of the surrounding off-site sources are hydraulically upstream of the site. The majority of the site is historically undeveloped.	Low	-
	Direct uptake	Flora	Site is situated on both a Non Aquifer and a Minor Aquifer. Groundwater flow into site is a very low possibility however some of the surrounding off-site sources are hydraulically upstream of the site. The majority of the site is historically undeveloped.	Low	-
	Direct contact	Building materials	Site is situated on both a Non Aquifer and a Minor Aquifer. Groundwater flow into site is a very low possibility however some of the surrounding off-site sources are hydraulically upstream of the site. The majority of the site is historically undeveloped.	Low	-
Elevated gas **On-site:** • Landfill **Off-site:** • None	Vertical & lateral migration	Site workers & occupiers	The majority of the site is historically undeveloped. Site is situated on both a Non Aquifer and a Minor Aquifer. Historic landfill could pose a minor contamination threat.	Low	-
		Adjacent occupiers	The majority of the site is historically undeveloped. Site is situated on both a Non Aquifer and a Minor Aquifer. Historic landfill could pose a minor contamination threat.	Low	-

Table 10c – Site SPR summary

11 On Site Landfill Investigation

11.1 During the preparation of geo-environmental phase 1 desk study, statutory registers identified the presence of a historical landfill site with a recorded coverage area as shown in Figure 11a below:

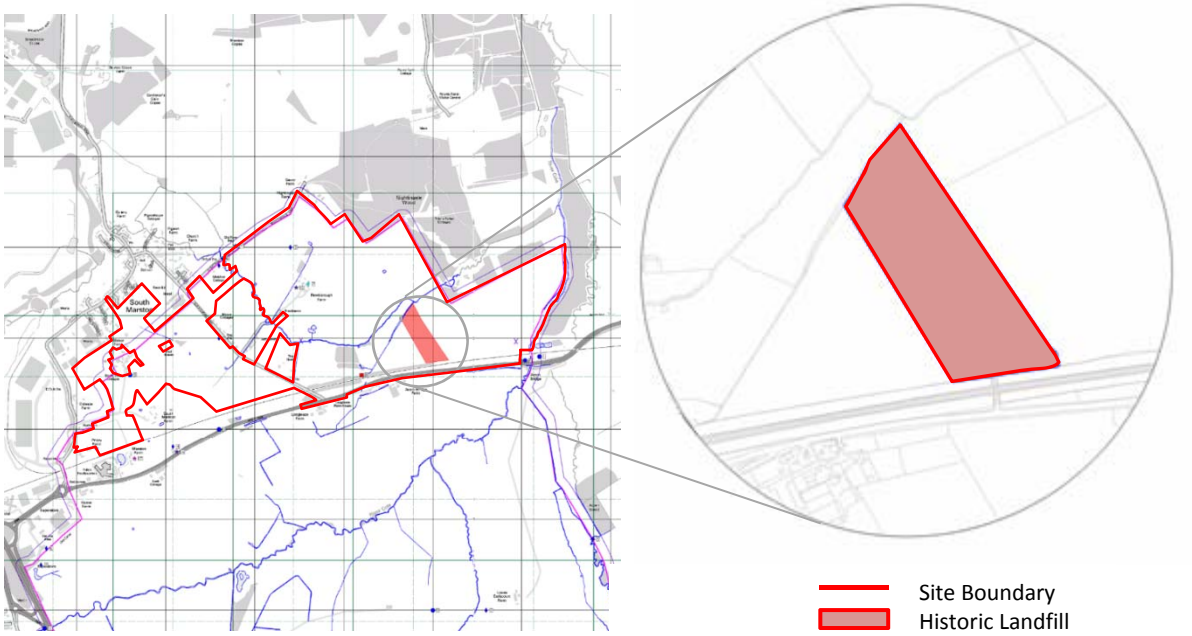


Figure 11a: Reported Landfill Coverage Area

11.2 Whilst identifying the presence of the historical landfill, statutory registers failed to adequately inform of the size, depth and composition of the material deposited in the landfill, although the deposited material is reported to be inert. Consequently, the potential contaminative risk presented to the proposed development of the site could not be appropriately determined and so intrusive investigations were undertaken to provide the necessary information.

11.3 Anecdotal evidence provided by land owner at the time the landfill was operational suggests that deposited material would comprise of inert construction waste originating from the development of Dorcan, Swindon, in the 1970's/1980's. The landfill was thought to be relatively shallow at around 1m deep with waste materials being deposited onto the land at the existing level.

11.4 In order to provide a robust assessment of the potential risks from the landfill within the site boundary, a Preliminary Phase II Site Investigation has been completed. This across the Historical Landfill area in December 2008 to inform of the geotechnical and contaminative conditions relative to the proposed development and also to determine the size and depth of the landfill and the nature of the waste material deposited. The factual report presents the findings of the investigation, this has been included within the appendix however, details of the works have been provided below.

Site Investigation

11.5 Site investigation works commenced Friday 12th December 2008, comprising of one day trial pitting and three cable percussion boreholes. The main focus of the trial pitting was to determine the horizontal and vertical extents of the landfill, to confirm the nature of the deposited material and allow collection of material for chemical analysis. The purpose of the boreholes was also to determine the vertical extents of the fill material but to also inform of the nature of the natural material below the landfill and to allow the installation of standpipes to investigate the potential for landfill ground gases.

11.6 During the investigation a local resident, who lived in close proximity to the site, approached to enquire about the works taking place. Through conversation with the local it became apparent he had good knowledge of the site, and he was able to inform of the coverage extent of the historical land fill and also historical land use at the site. It appeared that the

information provided by the statutory registers did not fully inform of the landfill coverage area and so the investigation was extended to the field boundaries based on the evidence provided by the local resident.

11.7 Subsequent investigation identified the land fill to cover an area of circa 9.7ha to the full extents of the field, which can be observed in Figure 11b below:

Figure 11b: Identified Landfill Coverage Area

Geology

11.8 All exploratory holes with the exception of TP6 were taken through the historical landfill coverage area. Consequently, in the reporting of the depth to geological formations, the strata encountered in TP6 will be reported separately.

11.9 Geology across the landfill coverage area can generally be characterised as:

- **Topsoil/MG:** Grass over soft to firm orange-brown slightly sandy clay of variable thickness ranging 150mm to 600mm across the site.
- **Made Ground:** Variable strength and stiffness, slightly sandy, slightly gravely grey clay with various fragments including, wood, metal, plastic, brick, concrete, wire and porcelain. (Upper Layers)
  Brown, grey and black slightly silty, slightly gravelly clays containing large fragments including wood, metal, plastic, brick, concrete. (Lower Layers)
  Depth of made ground encountered ranged 0.15m to 4.90m bgl.
- **Drift Deposits:** Generally in agreement with published geology the intrusive investigations did not encounter superficial deposits across the site.
- **Solid Deposits:** Soft/Firm/Stiff orange-brown sandy clay at depths to upper surface ranging 1.90m to 4.90m bgl.
  Moderately strong, red, orange and brown sandstone seams encountered at various depths.
  Firm to stiff high strength dark blue-grey clay at depths ranging 4.40m to 10.00m bgl.
  Strong to very strong white limestone at a depth of 7.70m bgl to upper surface.

11.10 The geology may be summarised as follows:

Stratum	Depth from surface	Thickness	Comments
Topsoil/MG	Surface	0.15m – 0.60m	Encountered in all exploratory holes
Made Ground	0.15m – 0.60m	2.20m – 4.50m	Encountered in all exploratory holes
Clay	1.90m – 4.90m	1.00m – 1.30m	Encountered in all exploratory holes, maximum thickness not proven
Sandstone	3.60m – 5.90m	0.20m – 0.30m	Encountered in BH1 & BH3 only.
Clay	3.90m – 6.10m	3.60m – 6.10m	Encountered in boreholes only, depth not proven
Limestone	7.70m	0.44m	Encountered in BH2 only, depth not proven

Figure 11c: Summary of Geology (TP6 excluded)

11.11 Trial pit TP6 was excavated on the field adjacent to landfill site which had a surface level circa 3m lower than the landfill site. The geology encountered can characterised as follows; topsoil comprising organic clay (surface to 0.15m bgl), overlying soft firm grey, orange, brown clay (0.15m to 1.50m bgl), over dark brown very clayey sand (1.50m to 2.00m bgl), over firm to stiff dark blue grey clay at 2m bgl (depth not proven).

11.12 The landfill material encountered appeared to be consistent with inert construction waste generally comprising of the following:

- Brick
- Mass Concrete
- Reinforced Concrete
- Precast Concrete (kerbs & paving slabs)
- Stone
- Tarmac
- Timber
- Plastic sheeting
- Plastic ties
- Polystyrene
- Electrical wiring
- Builders sand
- Ceramics
- Reworked natural material

11.13 Preliminary calculations based on site observations and exploratory hole logs, estimate the volume of made ground (from surface to top of natural strata) to be in the region of 300,000m^2 to 350,000m^2. An illustrative contour drawing of made ground thickness is shown below in figure 4a.

Historic Landfill Boundary

11.14 In a number of the trial pits organic odours could be identified, potentially indicating the biodegradation of some land fill materials which could result in the build-up of ground gases.

11.15 Groundwater was encountered in almost all exploratory holes at depths ranging 1.00m to 4.20m bgl. In the groundwater of trial pits excavated in the area of the former banger race track oils could be observed floating on the surface which suggests the possibility of hydrocarbon contamination.

11.16 Subsequent monitoring of three borehole installations identified water levels ranging 0.80m to 4.44m bgl, with fluctuations in level between visits ranging from 20mm to 1000mm.

11.17 During the investigation organic odours were observed in a number of trial pits. Subsequent monitoring of three borehole installation has identified elevated levels of carbon dioxide and methane. A summary of the gas monitoring levels is attached.

Chemical Contamination

11.18 The Environmental Protection Act 1990: Part 2A Contaminated Land Statutory Guidance was updated in April 2012; the guidance provides a definition of 'Contaminated Land' and sets out the nature of liabilities that can be incurred by owners of contaminated land and groundwater.

11.19 The Act defines contaminated land as:

11.20 "Contaminated land is any land which appears to the local authority in whose area it is situated to be in such a condition, by reason of substances in, on or under the land that – (a) significant harm is being caused or there is a significant possibility of such harm being caused; or (b) significant pollution of controlled waters is being caused, or there is a significant possibility of such pollution being caused;"

11.21 However, central to this definition, the Act provides further clarity with relation to land contamination:

"Contaminated land is used to mean land which meets the Part 2A definition of contaminated land. Other terms, such as "land affected by contamination" or "land contamination", are used to describe the much broader categories of land where contaminants are present but usually not at a sufficient level of risk to be contaminated land."

11.22 To this end, the guidance confirms that whilst contaminants may be present within the underlying strata, it may be the case that the land is not classified as "Contaminated Land" under the guidance

Risk Assessment Basis

11.23 Under Part 2A guidance, for a risk to exist there needs to be one or more contaminant-pathway-receptor linkages by which a relevant receptor might be affected by the contaminants in question. Therefore, there must be contaminants present in the land in a form or quantity that poses a hazard, and one or more pathways by which they might significantly harm people, the environment, or property; or significantly pollute controlled waters.

a) A "contaminant" is a substance which is in, on or under the land and which has the potential to cause significant harm to a relevant receptor, or to cause significant pollution of controlled waters.

b) A "receptor" is something that could be adversely affected by a contaminant, for example a person, an organism, an ecosystem, property, or controlled waters. The various types of receptors that are relevant under the Part 2A regime are explained in later sections.

c) A "pathway" is a route by which a receptor is or might be affected by a contaminant

11.24 In 2004 the Environment Agency published the 'Model Procedures for the Management of Land Contamination', CLR11, which provides the technical framework for applying a risk management process, using the Source-Pathway-Receptor model when dealing with land affected by contamination.

11.25 In 2008, to enable the practical application of good practice of the EA's Model Procedures CLR11, R&D Publication 66 'Guidance for the Safe Development of Housing on Land Affected by Contamination' (R&D66) was published by the National House Builders Council (NHBC), the EA and the Chartered Institute of Environmental Health. Whilst written to be relevant to housing development it is also applicable to other forms of development where sites are land affected by contamination. The guidance describes in detail the procedures and activities involved for the phased process of the management of land affected by contamination, covering hazard identification and assessment (Phase 1), risk estimation and evaluation (Phase 2), and remediation; design, implementation and verification (Phase 3).

11.26 As such, the assessment of risk within this report has been progressed in accordance with the guidance of R&D66.

11.27 Brookbanks Consulting approach the assessment of potentially contaminated land in line with the regulations, based on a tiered risk assessment methodology as follows.

Tier 1:

- Qualitative 'Source - Pathway - Receptor (target)' risk assessment to identify the 'pollutant linkages' of most concern.
- Screening of chemical test results against published guidelines to identify issues that may require further consideration.

Tier 2:

- Application of site specific quantitative risk assessment to further assess risks of concern. This may involve quantitative consideration of potential site specific exposure scenarios for contaminated soils, taking into account toxicological properties of substances to derive site specific target levels (SSTLs). Alternatively analytical calculations of groundwater flow and contaminant attenuation may be completed to ascertain if a risk is posed to a groundwater receptor and to allow acceptable levels at the site to be defined.

Tier 3:

- More complex modelling of contaminant behaviour in soils and / or groundwater.

11.28 Generally, increased tiers result in less conservative remediation target concentrations, which may result in less costly site remediation. However, the additional time and expense in completing further assessment needs to be considered in the context of the overall site objectives.

11.29 This report is primarily based upon a Tier 1 assessment, although screening criteria is augmented with Tier 2 assessments where appropriate.

Assessment Criteria

11.30 The assessment of contaminated land in the UK was historically undertaken with reference to ICRCL Guidance Note 59/83. The ICRCL Guidance provided Threshold and Action Trigger Concentrations (TTC's and ATC's) for a number of chemical determinants. The TTC is defined as a level where the risk to end users is no greater than is normally accepted and hence the soil sampled may be treated' as if it is uncontaminated. Where the ATC is exceeded, the risk associated by the presence of a specific determinant is deemed unacceptable, and the material must be treated as if it were contaminated. Where the concentration of a specific determinant is between the TTC and ATC the significance of the risk depends on the intended use and form of development, with professional judgement being required to decide whether any remedial action is needed.

11.31 The ICRCL guidance has now been formally withdrawn by DEFRA, and replaced by guidance which supports the implementation of the new Contaminated Land regime in the form of the Contaminated Land Exposure Assessment (CLEA) model. This was published by DEFRA and the Environment Agency on 14th March 2002.

11.32 The CLEA model is a probabilistic quantitative risk assessment (QRA) model that will be used to derive Soil Guideline Values (SGV's) for approximately 55 contaminants. The SGV's will be interpreted as generic intervention (action) values and will replace former ICRCL threshold and action concentrations. Currently, the CLEA derived SGVs have been published for the following eleven contaminants; arsenic, nickel, mercury, selenium, cadmium, benzene, toluene, ethylbenzene, xylene, phenol and dioxins, furans and dioxin-like PCBs, with additional guidelines due to be published as part of a future rolling programme. The CLEA model introduces detailed statistical analysis of contamination test data with the concept of the Mean Value Test (T_{mean}) and the Maximum Value Test (T_{max}) being critical to the understanding of data interpretation.

11.33 It should be noted that the CLEA model is generally considered to be mandatory with regards to Part IIA of the Environmental Protection Act 1990.

11.34 In 2006 Defra published a paper called "Soil Guideline Values: the Way Forward" which discussed various ideas for how the CLEA technical guidance might be revised to become more useful for assessors conducting risk assessments, and to help decide whether land qualifies as contaminated land under Part 2A of the EPA90. Subsequently, a list of measures to improve the guidance is to be implemented and reported to include; the revision guidance on how to assess the toxicology of substances and guidance setting out a methodology for estimating human exposure to chemicals in soil; the withdrawal of all the previous CLR7 – CLR10 documents; the release of a new version of the CLEA software, a technical review of body weight and height data used within the CLEA model and a technical review of physical-chemical properties for use within the CLEA model for the derivation of soil guidance values; an update of existing SGV's and the production of new guidance values; the development of other supporting information by the EA and other organisations; and the publication of new guidance on the use of statistics.

11.35 These reports have now been published along with "the Outcome of the Way Forward Report", DEFRA 2008. The two reports are titled "Human Health; Toxicological Assessment of Contaminants in Soil" and "Updated Technical Background to CLEA model".

11.36 The existing SGV guidance is being withdrawn by DEFRA and the EA and, at present, is being replaced by a series of new documentation, which is currently incomplete. Consequently, there is currently insufficient information available to complete assessments under the revised guidance. Given this status, it is therefore considered most appropriate to employ the former CLEA guidance for the purposes of this assessment.

11.37 In the absence of appropriate Tier 1 determinant screening criteria, issued as part of the CLEA framework, appropriate other generally accepted published guidelines have been adopted for soil contamination in the form of the Soil Code for phytotoxic metals.

11.38 Coupled with this, DEFRA and the BGS published guidance in 2012 called *"Normal Background Concentrations (NBC's) of Contaminants in English Soils"*. This document has been produced as a response to the following paragraphs taken from the revised Part 2A guidance which states:

"Normal levels of contaminants in soil should not be considered to cause land to qualify as contaminated land, unless there is a particular reason to consider otherwise. Therefore, if it is established that land is at or close to normal levels of particular contaminants, it should usually not be considered further.

11.39 *For the purpose of this Guidance, "normal" levels of contaminants in soil may result from:*

(a) The natural presence of contaminants (e.g. caused by soil formation processes and underlying geology) at levels that might reasonably be considered typical in a given area and have not been shown to pose an unacceptable risk to health or the environment.

(b) *The presence of contaminants caused by low level diffuse pollution, and common human activity other than specific industrial processes. For example, this would include diffuse pollution caused by historic use of leaded petrol and the presence of benzo[a]pyrene from vehicle exhausts, and the spreading of domestic ash in gardens at levels that might reasonably be considered typical.*

In deciding whether land has normal levels of contaminants, the local authority should consider whether contamination is within the bounds of what might be considered typical or widespread: (a) locally, if there is sufficient information to make a reasonable consideration of what is normal within a local area; and/or (b) regionally or nationally in broadly similar circumstances, having due regard to similarity in terms of land use and other relevant factors such as soil type, hydrogeology, and the form of the contaminants.

The local authority should decide that normal levels of contaminants exist in relation to land where: (a) those levels are not significantly different to those likely to be typical or widespread within the authority's area, or in other similar areas; and/or (b) those levels are common or usual in similar land use situations across England or parts of it; and (c) there is no specific reason to consider that those levels of contaminants are likely to pose an unacceptable risk.

11.40 This guidance provided a series of technical documents outlining the agreed NBC's for potential contaminants such as Arsenic and Nickel which have been used to define the classification of the site with regards to the contaminated land definition taken from the Part 2A guidance.

Soil Contamination

11.41 The soil screening appraisal, for residential end use, has identified two elevated contaminant levels. Both elevated levels were found in a madeground sample taken at a depth of 2.0m bgl from TP3.

11.42 Arsenic, measured at 34mg/kg, was identified to be marginally above the BGS Normal Background Concentration of 32mg/kg.

11.43 Chromium was found to be 11.8mg/kg above the derived site specific screening criteria of 23.2mg/kg.

11.44 The results of the tests and screening assessment are contained as Figure R1 which is attached.

Gas Contamination

11.45 Figure 6b below summarises the gas monitoring results:

Location	Pressure (mbar)	CH_4 (%)	CO ppm	O_2	CO_2	H_2S ppm	Flow (l/hr)
BH1	993	0.10	0.00	19.10	0.40	0.00	0.00
	1015	0.10	0.00	19.20	0.30	0.00	0.00
BH2	993	3.0	0.00	3.50	13.10	0.00	0.00
	1015	0.10	0.00	18.40	0.60	0.00	0.00
BH3	993	0.10	0.00	19.80	0.40	0.00	-0.10
	1015	0.10	0.00	19.90	0.10	0.00	0.00

Figure 11e: Gas Monitoring Results

11.46 The figure above indicates elevated levels of carbon dioxide and methane which provides some evidence of gassing from the landfill. Only a limited amount of monitoring has been completed therefore it is recommended that further monitoring is to be completed.

Site Development

11.47 General interpretation of the ground conditions indicates:

- Soil Penetration Tests (SPT's) completed in the natural strata of the exploratory boreholes at depths ranging 3 to 10m, identified a range of soil strengths varying with composition. SPT's taken in the orange-brown clays suggested allowable bearing pressures in the region of 150 to 200 kN/m^2, whereas those taken in the blue grey clay suggested allowable bearing pressures in the region of 200 to 250 kN/m^2. SPT's in sandstones indicated an allowable bearing pressure of circa 200 kN/m^2, whereas the SPT carried out in the limestone of BH2 indicated an allowable bearing pressure in excess of 500 kN/m^2. At this stage, the reported allowable bearing pressures are for preliminary indicative purposes. Prior to development, further investigation and assessment is recommended to accurately define the soil strength characteristics.

- CBR testing was not part of the site investigation scope and so no testing was completed. However, preliminary estimations based on hand shear vane tests completed in the madeground suggest CBR's are likely to be in the order 2-3%. For the purposes of design it is recommended CBR tests are completed to adequately inform of an appropriate value.

- At this stage, analysis of sulphate and magnesium levels suggests a likelihood of a Design Sulphate Class DS-3 and an Aggressive Chemical Environment for Concrete (ACEC) classification of AC-3 would be required.

- The identification of elevated gases in one of the monitoring installations suggest the site potentially would fall within gas characteristic situation 4 which would require the use if active gas protection measures. However, at this stage, as only limited gas monitoring has been carried out, further monitoring an interpretation is required to adequately inform of the appropriate gas regime.

Site Reclamation

11.48 At this stage, without comprehensive interpretation of the investigatory results and site observations, it would be irresponsible to advise of the potential site contaminative risk, or of the necessary reclamation strategies required to make the site suitable for use.

11.49 The contamination appraisal for residential end use identified only two elevated contamination levels, which initially do not appear to present a significant contamination risk. However, based on site observations and given the limited amount of testing completed to date, is considered the is a reasonable chance that further investigation may identify additional elevated contaminants, which potentially could the increase contamination risk. Therefore, to provide a suitably robust contaminative risk assessment and advise of appropriate reclamation strategies it is recommended further investigation and testing is completed.

11.50 The contamination appraisal for commercial end use did not identify any elevated contaminant levels and so the initial outlook on contaminative risk (with respect to commercial development) appears to be low. Contaminant levels are generally well below the relevant screening criteria and so while it is possible further investigation and testing may identify elevated contaminants, the probability of identifying significant contamination is not considered to be high. However, to provide a suitably robust contaminative risk assessment and advise of appropriate reclamation strategies it is recommended further investigation and testing is completed.

11.51 Gas monitoring to date has identified elevated levels of carbon dioxide and methane which provides evidence of gassing from the landfill however, to determine the appropriate site gas regime further monitoring is necessary.

11.52 It is considered that for a residential development that some reclamation works will be necessary, with residential development potentially requiring more reclamation.

12 Discussion & Summary

12.1 A review of readily available site environmental data, including historical mapping and statutory registers does not identify any former land uses that are potentially contaminative or likely to be prohibitive to the planned development. The contaminative risk at the site is considered to be low.

12.2 **Rail Line:** Development of a residential nature surrounds the length of the line through Swindon and therefore it is considered that this feature provides a low rating for risk.

12.3 **Historic Landfill:** The site investigation identified the landfill to cover an area of circa 9.7 ha, to an average depth of around 3.0m below ground level. Deposited materials generally appeared to be consistent with inert construction waste, as the statutory registers and anecdotal evidence suggests, and preliminary calculations estimate the volume of made ground across the site to be in the region of 300,000 m^3 to 350,000 m^3.

12.4 With respect to residential development, chemical analysis of the madeground encountered across the site identified some minor contamination within the soils. It is anticipated there is a reasonable chance that further testing may identify additional elevated contaminants given observations made during intrusive investigations. Therefore, it is recommend further investigation, testing and interpretation is completed to allow the development of a robust contaminative risk assessment and reclamation strategy.

12.5 Geotechnical investigation and testing has identified the natural strata below the madeground to comprise predominantly of clays of reasonable strength, but also seams of sandstone and a formation of limestone. Preliminary interpretation suggests the natural strata beneath the madeground are generally appropriate for receiving traditional and specialist, residential foundations. However, the variation in the material and depth of the overlying madeground presents a number of complications to the development of a suitable foundation strategy. As such, further interpretation of the site conditions is deemed necessary before a strategy can be proposed.

12.6 Gas monitoring to date has identified elevated levels of carbon dioxide and methane which provides some evidence of gassing from the landfill. However, at this stage only a limited amount of monitoring has been completed and so prior to the recommendation of an appropriate gas regime it is considered that further monitoring is necessary.

12.7 Potential contaminative risks have been identified from the desk based inspections however, these are unlikely to be prohibitive to development and further investigations will be required.

13 Limitations

12.1 The benefits of this report are provided solely to Swindon EDA Consortium. The conclusions and recommendations contained herein are limited to those given the general availability of background information and the planned usage of the site. Brookbanks Consulting do not confer any third party rights for the information contained in the report.

12.2 All distances referred to in this report are measured from the boundary of the planned development site unless otherwise advised.

12.3 Third party information has been used in the preparation of this report, which Brookbanks Consulting, by necessity assume is correct at the time of writing.